How-To Library

Informatica

Getting Started with Informatica Intelligent Cloud Services and Databricks Delta

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Introduction

This guide will help you navigate the Informatica Intelligent Cloud Services℠ (IICS) interface and to load data into Databricks Delta quickly.

This guide contains two sections. The first section describes the initial steps for setting up your organization to use Databricks Delta. It explains how to perform the following tasks:

- Create an IICS organization.
- Install a Secure Agent.
- Create connections to a Databricks Delta SQL warehouse.

The second section of this guide includes examples of commonly used features for loading data into Databricks Delta. It explains how to perform the following tasks:

- Load a CSV file into Databricks Delta using a mapping.
- Load an Amazon S3 file into Databricks Delta using a data loader task.
- Use the SQL ELT optimization option in a mapping task.

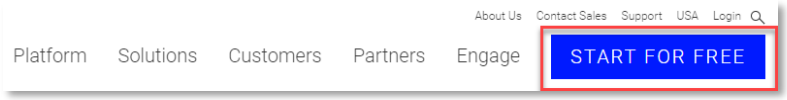
Section 1: Setting up your organization

Creating an IICS organization

You can skip this step if you already have an IICS organization with Informatica.

To create an IICS organization, perform the following steps:

1. Visit the Informatica home page at www.informatica.com.
2. Click **Start for Free**.



3. Click **Get Started for Free**.
4. Complete the sign-up form.

   Informatica will contact you to complete the organization setup process.

Installing a Secure Agent

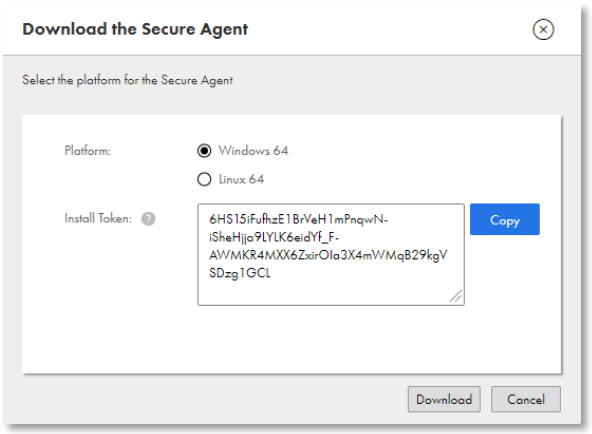
You can skip this step if you already have an IICS organization with one or more Secure Agents installed.

The Secure Agent is a small footprint application that enables secure communication across the firewall between your organization and IICS. It enables IICS to get through the firewall to access application, relational database, and file sources and targets in your on-premises network.

Perform the following steps to install the Secure Agent on a server in your on-premises network or in the cloud service provider of your choice:

1. In the Administrator service, click **Runtime Environments**.

2. Click Download Secure Agent.
3. Select the operating system platform where you will install the Secure Agent.



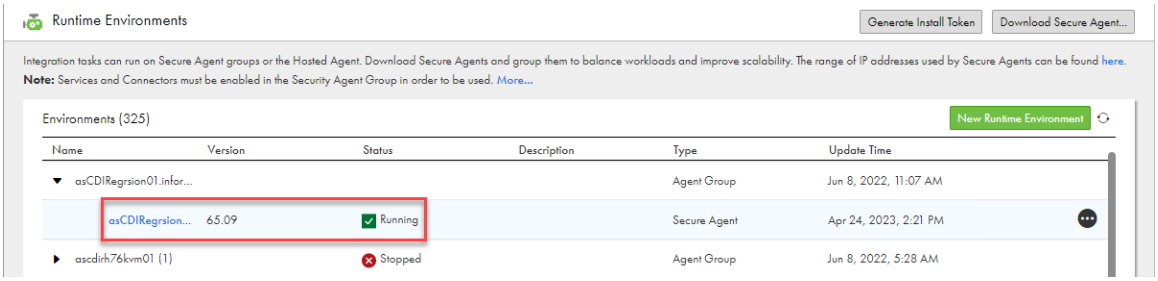
4. Click **Copy** to copy the install token string.

   The install token is required to associate the Secure Agent with your IICS organization.

5. Click **Download** to start downloading the Secure Agent.
6. Install the Secure Agent and supply the install token when requested.

   For more information about deploying a Secure Agent, see these instructions in the IICS *Administrator* guide or check out this video.

7. Verify that the Secure Agent is listed on the **Runtime Environments** page in Administrator:



Note: Your runtime environment should be listed with the status “Running” before you continue to the next step. If you see the status “Not all the services are running,” wait 5-10 minutes for the agent to start up completely and install updates.

Creating connections to a Databricks Delta SQL warehouse

The Databricks Delta connection uses the SQL warehouse method. You create a Databricks Delta connection so that you can either read data from or write data to Databricks Delta. Once you’ve created a Databricks Delta connection, you can use this connection in the Source or Target transformations in mappings and mapping tasks.

Tip: The connections properties described in this document also apply to other services such as:

- Cloud Data Integration

- Data Integration-Free & PayGo
- Data Loader
- Cloud Mass Ingestion

  For more information about using Databricks Delta targets in Mass Ingestion, see "Databricks Delta targets" in the Informatica documentation portal.

To create a Databricks Delta connection to a Databricks SQL Warehouse, perform the following steps:

1. Configure Spark parameters for the SQL warehouse.
2. Configure environment variables for the SQL warehouse.
3. Create a minimal Amazon IAM policy.
4. Configure role-based access control for Microsoft Azure Data Lake Storage Gen2.
5. Configure the Databricks Delta connection properties.

Step 1. Configure Spark parameters for the SQL warehouse

Before you use the Databricks SQL warehouse to run mappings, configure the Spark parameters for the SQL warehouse.

1. Start the Databricks SQL Admin console.
2. Navigate to **SQL Warehouse Settings > Data Security**.
3. Under **Data access** configuration, configure the Spark parameters for AWS or Azure.

AWS	Add the following Spark configuration parameters and restart the SQL warehouse:
	`spark.hadoop.fs.s3a.access.key <S3 access key value>`
	`spark.hadoop.fs.s3a.secret.key <S3 secret key value>`
	`spark.hadoop.fs.s3a.endpoint <S3 staging bucket endpoint value>`
	An example of the S3 staging bucket endpoint value: **s3.ap-south-1.amazonaws.com**
	Ensure that the access and secret key configured has access to the buckets where you store the data for Databricks Delta tables.

Azure	Add the following Spark configuration parameters and restart the SQL warehouse:
	`spark.hadoop.fs.azure.account.oauth2.client.id.<storage account name>.dfs.core.windows.net <value>`
	`spark.hadoop.fs.azure.account.auth.type.<storage account name>.dfs.core.windows.net OAuth`
	`spark.hadoop.fs.azure.account.oauth2.client.secret.<storage account name>.dfs.core.windows.net <Value>`
	`spark.hadoop.fs.azure.account.oauth.provider.type.<storage account name>.dfs.core.windows.net org.apache.hadoop.fs.azurebfs.oauth2.ClientCredsTokenProvider`
	`spark. hadoop.fs.azure.account.oauth2.client.endpoint.<storage account name>.dfs.core.windows.net https://login.microsoftonline.com/<Tenant ID>/oauth2/token`
	Ensure that the client ID and client secret configured have access to the file systems where you store the data for Databricks Delta tables.

Step 2. Configure environment variables for the SQL warehouse

Set the following environment variables in the Secure Agent before you connect to the Databricks SQL warehouse:

```
export LANGUAGE="en_US.UTF-8"
export LC_ALL="en_US.UTF-8"
```

After you set the environment variables, restart the Secure Agent.

Step 3: Create a minimal Amazon IAM policy

Create an Amazon Identity and Access Management (IAM) policy and define the required permissions to stage the data in Amazon S3.

Use the following minimum required permissions to stage the data in Amazon S3:

- PutObject
- GetObject
- DeleteObject
- ListBucket
- ListBucketMultipartUploads. Applicable only for mappings in advanced mode.

You can use the following sample Amazon IAM policy:

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "s3:PutObject",
                "s3:GetObject",
                "s3:DeleteObject",
                "s3:ListBucket",
                "s3:ListBucketMultipartUploads"
                        ],
            "Resource": [
                "arn:aws:s3:::<bucket_name>/*",
                "arn:aws:s3:::<bucket_name>"
            ]
        }
    ]
}
```

For mappings in advanced mode, you can use different AWS accounts within the same AWS region. Make sure that the Amazon IAM policy confirms access to the AWS accounts used in these mappings.

Note: Test Connection does not validate the IAM policy assigned to users. The Amazon S3 bucket name is available in the source and target advanced properties.

Step 4. Configure role-based access control for Microsoft Azure Data Lake Storage Gen2

Before you use Microsoft Azure Data Lake Storage Gen2 to stage files, perform the following tasks:

- Create a storage account to use with Microsoft Azure Data Lake Storage Gen2 and enable hierarchical namespace in the Azure portal.

  You can use role-based access control to authorize the users to access the resources in the storage account. Assign the Contributor role or Reader role to the users. The Contributor role grants you full access to manage all resources in the storage account but does not allow you to assign roles. The Reader role allows you to view all resources in the storage account but does not allow you to make any changes.

  Note: To add or remove role assignments, you must have write and delete permissions. An example is the Owner role.

- Register an application in Azure Active Directory to authenticate users to access the Microsoft Azure Data Lake Storage Gen2 account.

  You can use role-based access control to authorize the application. Assign the Storage Blob Data Contributor or Storage Blob Data Reader role to the application. The Storage Blob Data Contributor role lets you read, write, and delete Azure Storage containers and blobs in the storage account. The Storage Blob Data Reader role lets you read and list Azure Storage containers and blobs in the storage account.

- Create an Azure Active Directory web application for service-to-service authentication with Microsoft Azure Data Lake Storage Gen2.

  Note: Ensure that you have superuser privileges to access the folders or files created in the application using the connector.

- To read and write complex files, set the JVM options in the Data Integration Service (DIS) for type DTM to increase the -Xms and -Xmx values in the system configuration details of the Secure Agent to avoid a Java heap space error. The recommended -Xms and -Xmx values are 512 MB and 1024 MB respectively.

Step 5. Configure the Databricks Delta connection properties

This configuration assumes that you are using the Unity catalog as the source.

To configure a Databricks Delta connection, complete the following tasks:

1. In Administrator, click **Connections**.
2. Click **New Connection** in the top right corner.
3. Enter a name and optional description for the connection.

4. Select **Databricks Delta** as the connection type.
5. Select the runtime environment that will run the tasks. Any runtime environment will work, but for this example, select the Secure Agent that you created earlier.
6. Configure the following connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: `. + - ,` Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Type	The Databricks Delta connection type.
Runtime Environment	Name of the runtime environment where you want to run the tasks. • *For Data Integration:* Select a Secure Agent, the Hosted Agent, or a serverless runtime environment. The Hosted Agent can't be used for mappings in advanced mode. • *For Data Loader:* Select the Informatica Cloud Hosted Agent. • *For Data Integration-Free and PayGo:* Select a Secure Agent or the Hosted Agent.
Databricks Token	Required. Personal access token to access Databricks Delta.

SQL Warehouse JDBC URL	Required. Databricks SQL warehouse JDBC connection URL.
	To get the SQL Warehouse JDBC URL, go to the Databricks Delta console and select the JDBC driver version 2.6.22 or later from the JDBC URL menu.
	Use the following syntax if you are using the Unity catalog:
	`jdbc:spark://<Databricks Host>:443/default;transportMode=http;ssl=1;AuthMech=3; httpPath=/sql/1.0/endpoints/<SQL warehouse cluster ID>;ConnCatalog=<catalog_name>;`
	Use the following syntax for all other instances:
	`jdbc:spark://<Databricks Host>:443/default;transportMode=http;ssl=1;AuthMech=3; httpPath=/sql/1.0/endpoints/<SQL warehouse cluster ID>;`

Property	Description
Cluster Environment	Required. The cloud provider where the Databricks Delta cluster is deployed. Choose from the following options: **AWS** or **Azure**. Default is **AWS**. The connection attributes depend on the cluster environment you select. A connection is tied to a cluster. If you change the cluster, ensure that the new cluster meets all the requirements. Cluster properties are required in the following scenarios: • When you create and run mappings to write data to Databricks Delta. • When you create and run mappings in advanced mode to read from and write to Databricks Delta. For more information, see AWS cluster properties and Azure cluster properties in this document.
Database	Optional. The database in Databricks Delta that you're connecting to. Specify a database name or specify **default** to enable the default database name. By default, all databases available in the workspace are listed.
JDBC Driver Class Name	Optional. Enter **com.simba.spark.jdbc.Driver** as the JDBC driver class name.

AWS cluster properties

When you set up a Databricks Delta connection, configure the connection properties based on the cluster environment you select.

The following table describes the Databricks Delta connection properties that apply when you select the AWS cluster environment:

Property	Description
S3 Access Key	The key to access the Amazon S3 bucket.
S3 Secret Key	The secret key to access the Amazon S3 bucket.
S3 Data Bucket	The existing bucket to store the Databricks Delta data.
S3 Staging Bucket +	The existing bucket to store staging files.
S3 Authentication Mode	The authentication mode to access Amazon S3. You can select one of the following authentication types: • **Permanent IAM credentials.** Uses the S3 access key and S3 secret key to connect to Databricks Delta.

Property	Description
	• **IAM Assume Role+.** Uses the AssumeRole for IAM authentication to connect to Databricks Delta. Before you select the IAM Assume Role, complete the steps described in the following topics in the Informatica documentation portal: • IAM Authentication • Temporary security credentials using AssumeRole.
IAM Role ARN +	The Amazon Resource Number (ARN) of the IAM role assumed by the user to use the dynamically generated temporary security credentials. Set the value of this property if you want to use the temporary security credentials to access the Amazon S3 staging bucket. For more information about how to get the ARN of the IAM role, see the AWS documentation. Applicable if you select IAM Assume Role as S3 Authentication Mode.
Use EC2 Role to Assume Role +	Optional. Select the check box to enable the EC2 role to assume another IAM role specified in the IAM Role ARN option. The EC2 role must have a policy attached with a permission to assume an IAM role from the same or different AWS account. Applicable if you select IAM Assume Role as S3 Authentication Mode.
S3 Region Name +	The AWS cluster region in which the bucket you want to access resides. Select a cluster region if you choose to provide a custom JDBC URL that does not contain a cluster region name in the JDBC URL connection property. Applicable if you select IAM Assume Role as S3 Authentication Mode.
S3 Service Regional Endpoint	The S3 regional endpoint when the S3 data bucket and the S3 staging bucket need to be accessed through a region-specific S3 regional endpoint. Default is **s3.amazonaws.com**
+Doesn't apply to mappings in advanced mode.	

Azure cluster properties

When you set up a Databricks Delta connection, configure the connection properties based on the cluster environment that you select.

The following table describes the Databricks Delta connection properties that apply when you select the Azure cluster environment:

Property	Description
ADLS Storage Account Name	The name of the Microsoft Azure Data Lake Storage account.
ADLS Client ID	The ID of your application to complete the OAuth Authentication in the Active Directory.
ADLS Client Secret	The client secret key to complete the OAuth Authentication in the Active Directory.
ADLS Tenant ID	The ID of the Microsoft Azure Data Lake Storage directory that you use to write data.
ADLS Endpoint	The OAuth 2.0 token endpoint from where authentication based on the client ID and client secret is completed.
ADLS Data Filesystem Name	The name of an existing file system to store the Databricks Delta data.
ADLS Staging Filesystem Name +	The name of an existing file system to store the staging data.
+Doesn't apply to mappings in advanced mode.	

Save and then test the connection. You should see the message, "The test for this connection was successful". If the test is not successful, review the properties and try again

For more information about configuring a Databricks Delta connection, see Databricks Delta connection properties in the Informatica documentation portal.

Section 2: Using Databricks Delta SQL ELT optimization

When you run a task configured for SQL ELT optimization, the task converts the transformation logic to an SQL query. The task sends the query to the database, and the database executes the query.

Having the database execute a SQL query is often faster than having the task interpret the transformations, thus improving overall task performance.

The amount of transformation logic that you can push to the database depends on the database, transformation logic, and task configuration. The Secure Agent processes all transformation logic that it cannot push to the database.

You can configure SQL ELT optimization for a mapping that contains a Databricks Delta connection. SQL ELT optimization enhances the mapping performance. You can configure full or source optimization when you read data from a Databricks Delta source and write to a Databricks Delta target.

Supported transformations for SQL ELT optimization

You can use full or source SQL ELT optimization to push the following transformations to Databricks Delta:

Aggregator
Expression
*Filter**
Joiner
*Lookup**
Sorter
Union
*Router***
*Rank**
*SQL**

*Doesn't apply to mappings in advanced mode.

**Router transformation is not applicable for source SQL ELT optimization.

Use case 1: Read from and write to Databricks Delta

You can configure SQL ELT optimization in a mapping to read from and write to Databricks Delta using a Databricks Delta connection.

Scenario:

You work in a motorbike retail company with more than 30,000 dealerships and 2000 inspection centers globally. The company stores millions of records in Databricks Delta hosted on Azure. You want to use Data Integration to perform some transformations on the data before you write back to Databricks Delta.

Use a Databricks Delta connection in the mapping to read from the Databricks Delta source and write the processed data to the Databricks Delta target. Configure full SQL ELT optimization in the mapping to enhance the performance.

Use case 2: Read from Amazon S3 and write to Databricks Delta

You can configure SQL ELT optimization for a mapping that uses an Amazon S3 V2 connection in the Source transformation to read from Amazon S3 and a Databricks Delta connection in the Target transformation to write to Databricks Delta.

Scenario:

You work for a healthcare organization. Your organization offers a suite of services to manage electronic medical records, patient engagement, telephonic health services, and care coordination services. The organization uses infrastructure based on Amazon Web Services and stores its data on Amazon S3. The management plans to load data to a data warehouse to perform healthcare analytics and create data points to improve operational efficiency. To load data from an Amazon S3-based storage object to Databricks Delta, you must use ETL and ELT with the required transformations that support the data warehouse model.

Use an Amazon S3 V2 connection to read data from a file object in an Amazon S3 source and a Databricks Delta connection to write to a Databricks Delta target. Configure full SQL ELT optimization in the mapping to optimize the performance.

Use case 3: Read from Microsoft Azure Data Lake Storage Gen2 and write to Databricks Delta

You can configure SQL ELT optimization for a mapping that uses a Microsoft Azure Data Lake Storage Gen2 connection in the Source transformation to read from Microsoft Azure Data Lake Storage Gen2 and a Databricks Delta connection in the Target transformation to write to Databricks Delta.

Scenario:

You want to load data from a Microsoft Azure Data Lake Storage Gen2-based storage object to Databricks Delta for analytical purposes. You want to transform the data before it is made available to users. Use a Microsoft Azure Data Lake Storage Gen2 connection to read data from a Microsoft Azure Data Lake Storage Gen2 source and a Databricks Delta connection to write to a Databricks Delta target. Configure full SQL ELT optimization in the mapping task to optimize the performance of loading data to Databricks Delta. SQL ELT optimization enhances the performance of the task and reduces the cost involved.

Using SQL ELT optimization in a mapping task

This section gives an example of a SQL ELT optimization scenario that uses an Amazon S3 source and a Databricks Delta target.

The example involves the following steps:

1. Create an Amazon S3 source connection.
2. Create a Databricks Delta target connection.
3. Create a mapping.
4. Create and run a mapping task.

Step 1. Create an Amazon S3 connection.

An Amazon S3 V2 connection lets you read data from or write data to Amazon S3. Use Amazon S3 V2 connections to specify sources and targets in mappings and mapping tasks.

To create an Amazon S3 connection, follow these steps:

1. Log in to Informatica Intelligent Cloud Services.
2. In Administrator, select **Connections**.
3. Click **New Connection**.
4. Enter a name and optionally a description for the connection.
5. Select **Amazon S3 v2 (Informatica)** for the connection type.
6. Complete the rest of the connection properties.

   For more information, see Amazon S3 V2 connection properties in the Informatica documentation portal.

7. To test the connection, click **Test Connection**. You should see a success message. If there are errors, check your properties and try again.
8. Click **Save**.

Step 2. Create a Databricks Delta target connection.

Create a Databricks Delta connection to connect to Databricks Delta and read data from Amazon S3 and write data to Databricks Delta. You can use Databricks Delta connections to specify sources or targets in mappings and mapping tasks.

To create a Databricks Delta connection, follow these steps:

1. In Administrator, select **Connections**.
2. Click **New Connection**.
3. Enter a name and optionally a description for the connection.
4. Select **Databricks Delta (Informatica Cloud)** for the connection type.
5. Complete the rest of the connection properties. For more information, see Connections for Databricks Delta in the Informatica documentation portal.
6. To test the connection, click **Test Connection**. You should see a success message. If there are errors, check your properties and try again.
7. Click **Save**.

Step 3. Create a mapping.

To create a new mapping:

1. Click **Administrator** next to the Informatica logo in the top left corner of the screen, and then select **Data Integration** from the menu.
2. Click **New** in the navigation menu on the left.
3. In the New Asset dialog box, select **Mappings**. The Mapping Designer creates a basic mapping with the two transformations that you need:

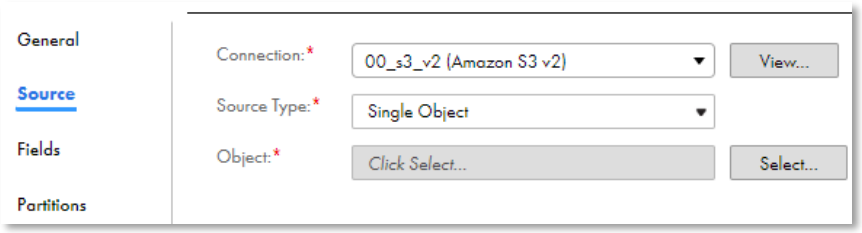


4. Configure the transformations as described in the following table:

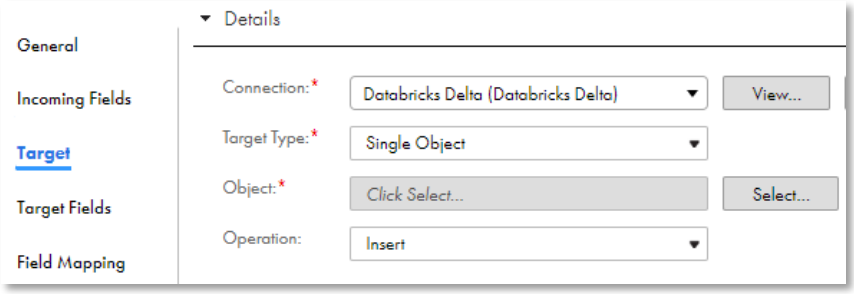
Transformation	Description	Configuration
Source	The S3 object that you read data from.	Source tab: • Select the Amazon S3 connection that you created earlier. • Select the S3 object that you are reading from. For more information, see Source transformation in the Informatica documentation portal.

Transformation	Description	Configuration
Target	The Databricks object that you want to write data to.	Target tab: • Select the Databricks Delta connection that you created earlier. • Browse and select the object or create a new target object based on your field projection. • Select the write action (insert, upsert etc.). Field Mapping tab: • If you use an existing Databricks object, configure the field mapping. For more information, see Target transformation in the Informatica documentation portal.

The following figure is an example of the Source transformation:



The following figure is an example of the Target transformation:



5. In the top right corner of the Mapping Designer, click to validate the mapping.

   If the mapping has validation errors, check the Data Integration online help for more information.

   The following are the most common validation errors:

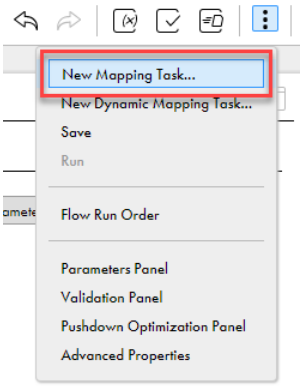
   - No source or target object selected.
   - Transformations not linked.

- No fields are mapped for an existing target.

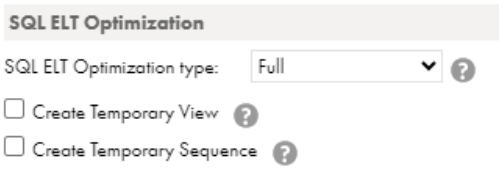
6. Click **Save** to save the mappings.

Step 4. Create and run a mapping task.

1. In the top right corner of the Mapping Designer, open the **Actions** menu and select **New Mapping Task**.



2. On the **Definition** page of the New Mapping Task wizard, enter a name for the mapping task, select the runtime environment, and then select the mapping that you created.
3. Click **Next** to reach the Schedule page.
4. On the Schedule page, scroll down to the **SQL ELT optimization** section.

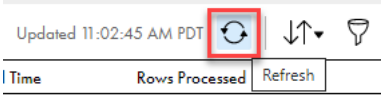


5. Select **Full** for the SQL ELT optimization type.
6. Click **Finish** to save and create the task. The task details are displayed, and the task name appears in the left panel.
7. Click **Run** to run the task.

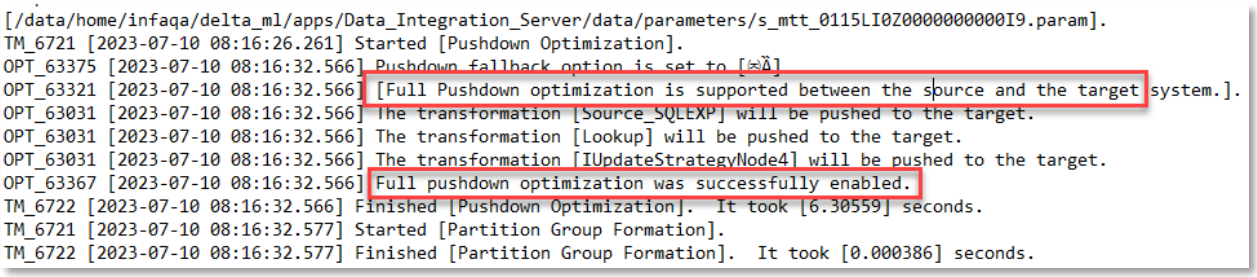


8. Click **My Jobs** to open the job activity page.

   It might take a few seconds for the new job to show up on the **My Jobs** page. Click refresh if you don't see it after a few seconds.



9. When processing is complete, check the PDO log to verify that the SQL ELT optimization was successful. You should see the following lines in the log:



Section 3: IICS and Databricks Delta ingest scenarios

Loading a CSV file to Databricks Delta using a mapping

1. Log in to Informatica Intelligent Cloud Services.
2. In Data Integration, select **New** in the navigation menu on the left.
3. In the **New Asset** dialog box, select **Mappings**.
4. Click **Create**. This creates a new mapping with empty Source and Target transformations.
5. Select the Source transformation and then edit its properties as described in the follow table:

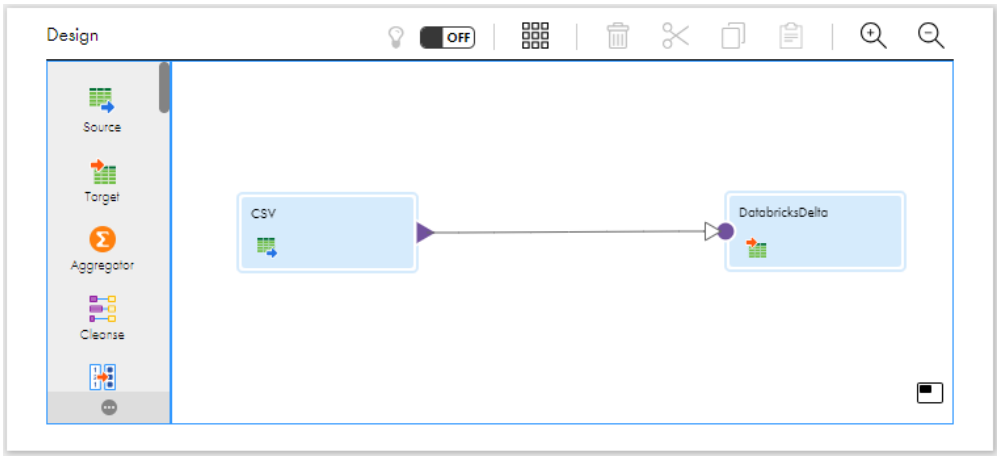
Property	Setting
General tab	
Name	CSV
Source tab	
Connection	Flatfile (Flat File)
Source Type	Single Object
Object	Click **Select** and then select your CSV file.

6. Select the Target transformation and then edit its properties as described in the follow table:

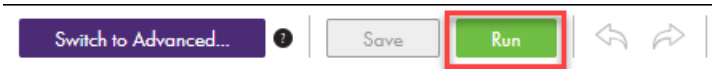
Property	Setting
General tab	
Name	DatabricksDelta
Source tab	
Connection	Databricks Delta (Databricks Delta)
Target Type	Single Object

Property	Setting
Object	1. Click **Select**. The Target Object dialog box is displayed. 2. Select **Create New at Runtime** for the Target Object. 3. Enter **CSV_TABLE** for the Object Name. 4. Click **OK**.
Operation	Insert

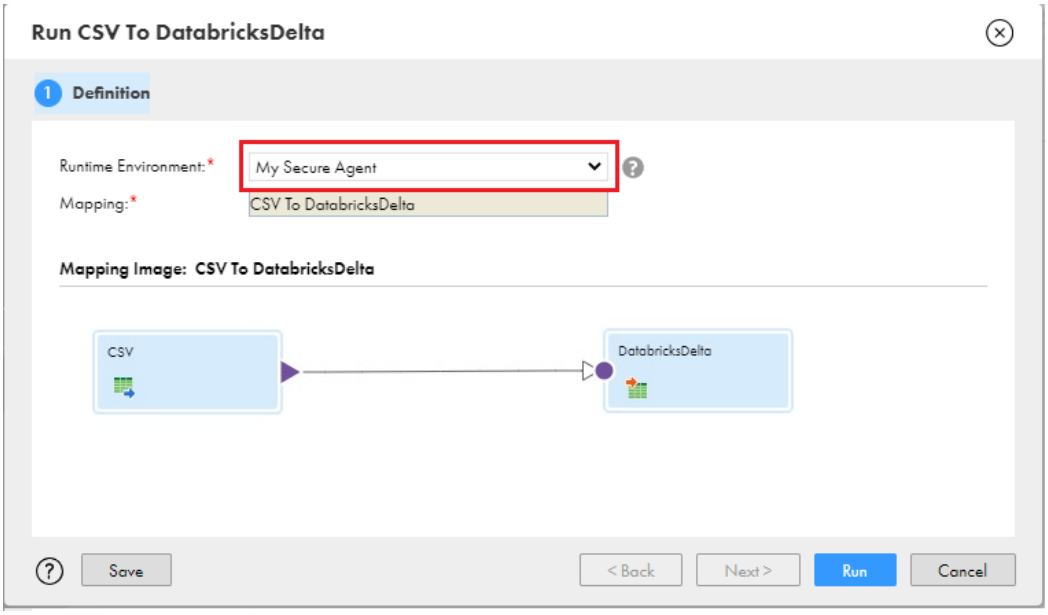
Your mapping should look like the following illustration:



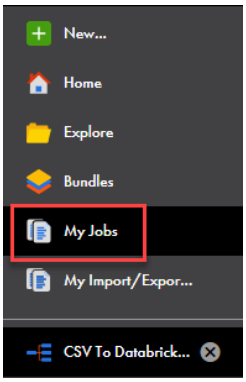
7. Now you can run your new mapping. Click **Run** the mapping designer:



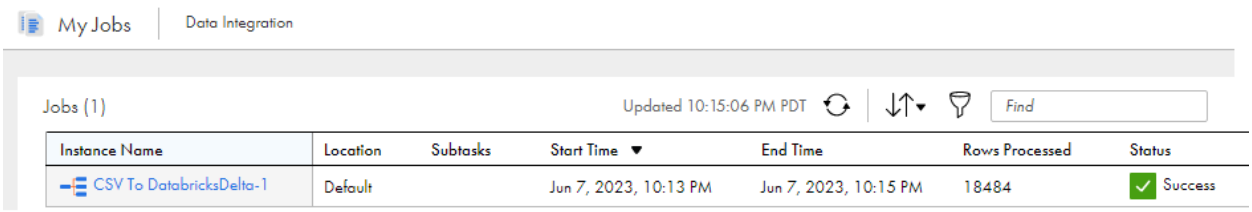
8. For the runtime environment, select the Secure Agent that you created earlier.



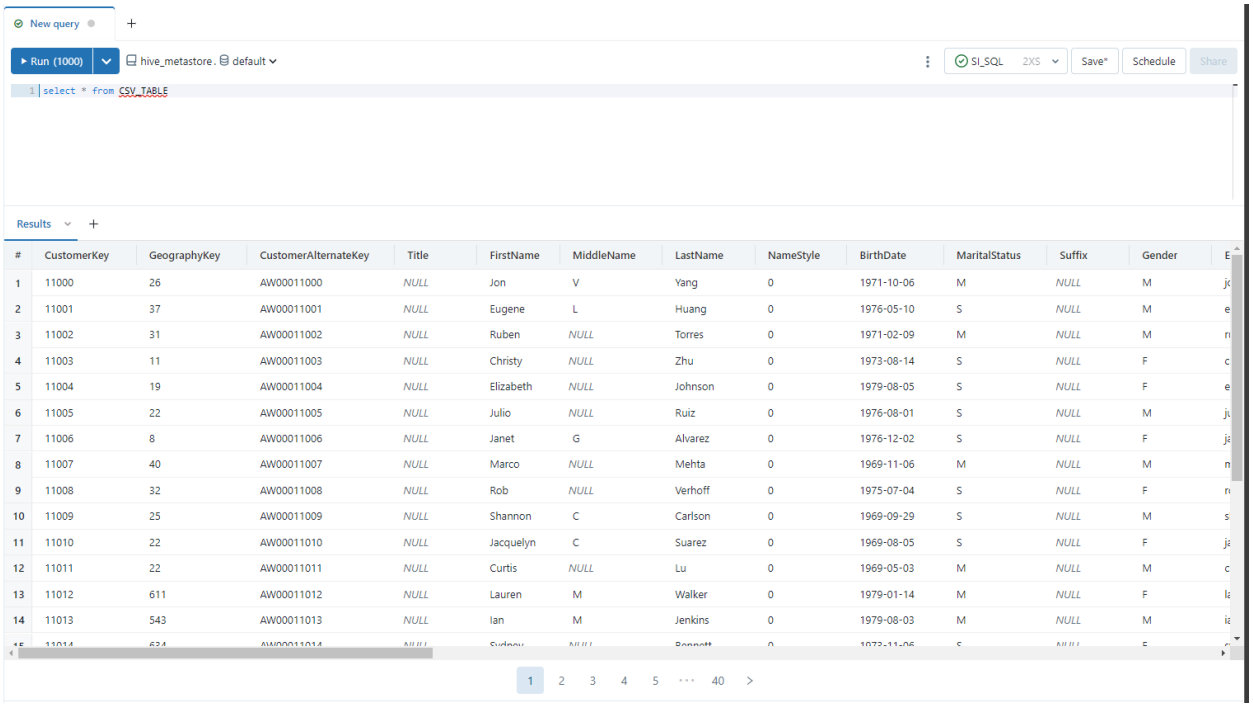
9. Click **Run**.
10. Click **My Jobs** from the left pane:



11. The My Jobs page shows the progress of your job. When it completes successfully, the status shows **Success**.



The following illustrations shows a sample inserted CSV file in the Databricks SQL Editor:



Loading an Amazon S3 file into Databricks Delta using a Data Loader task

Data Loader is a free product from Informatica that you can use to load large data sets to your cloud warehouse quickly and securely.

In this example, you will load data from Amazon S3 into Databricks Delta. (Flat files such as CSV are not currently supported by Data Loader with Databricks Delta as the target.)

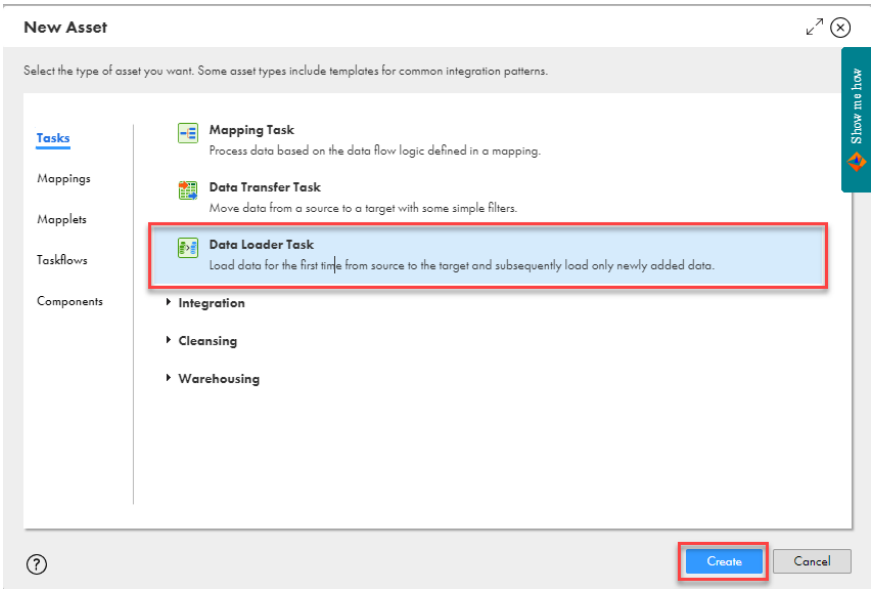
For more information, see Data Loader in the Information documentation portal.

Step 1. Connect to the source.

1. Sign up here to get started with Data Loader.
2. Log in to Informatica Intelligent Cloud Services.
3. In Data Integration, select **New** in the navigation menu on the left.

   If you see a new empty Data Loader task, proceed to the next step.

   If you see a New Asset dialog box, select **Data Loader Task,** and then click **Create**. This creates a new empty Data Loader task.



4. On the **Connect Source** page, click **New Connection**.
5. Select **Amazon S3 v2**.



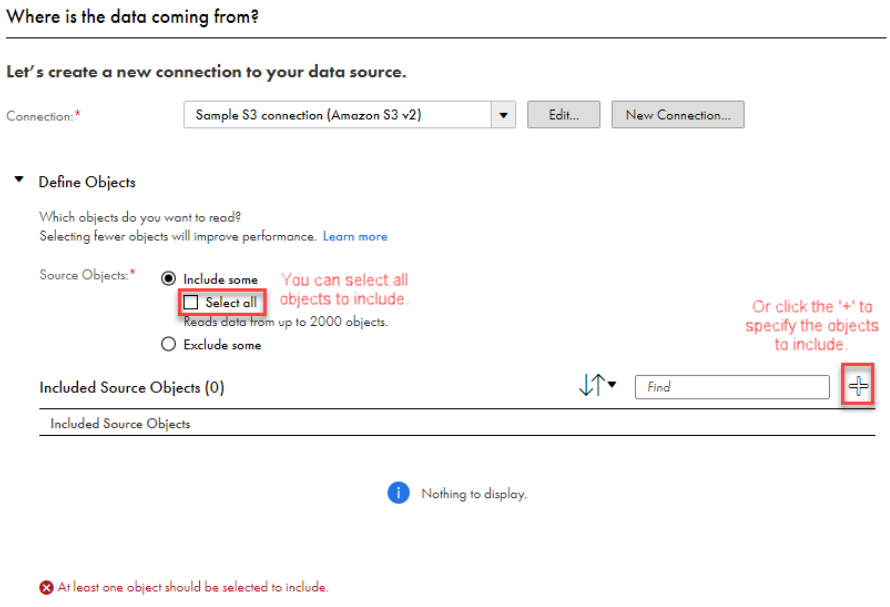
6. Click **Continue Setup**.

7. Complete basic information about your Amazon S3 connection, using the Setup Guide to assist you.



8. Click **Add**.
9. Configure the fields as shown in the following table:

Field	Setting
Connection	This is the Amazon S3 connection that you just configured. If the connection is incorrect, click **New Connection** and configure one now.
Define Objects	Select which source objects to include: some, all, or all except certain objects. Reducing the number of objects to process improves performance. You must select at least one object to include.



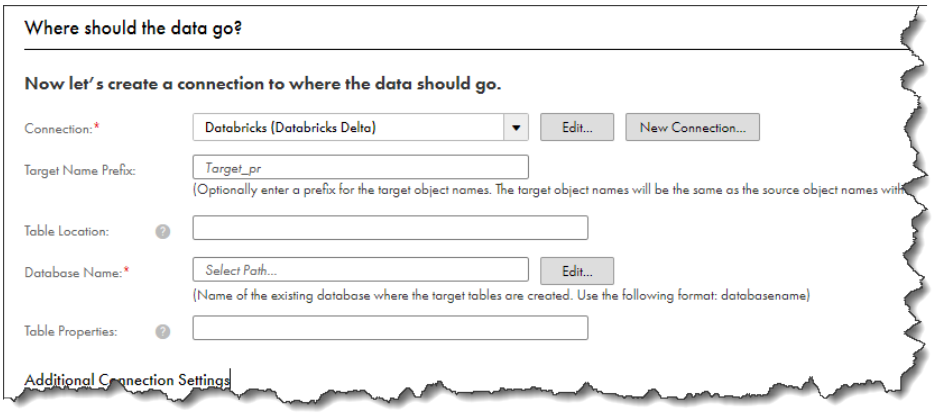
10. There are many controls at the bottom of the page for you to fine tune the source data. In this example, we'll skip the fine tuning. Click **Next>**.

Step 2. Connect to the target.

After you configure the source, select, and configure the target.

1. On the **Connect Target** page, configure the fields as shown in the following table:

Field	Setting
Connection	Select **Databricks (Databricks Delta)**.
Target Name Prefix	Optionally enter a prefix to add to the target object names. A prefix is helpful for identification purposes.
Database Name	Click **Edit** and select an existing database to create the target tables.



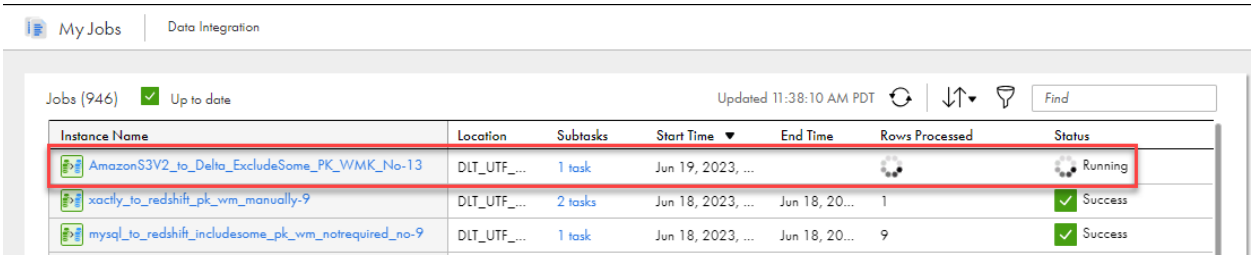
2. Leave all the other fields at their defaults. Click **Next>**.

Step 3. Run the task.

In this last step, you can customize your task by setting a schedule, email addresses for notifications, etc. For this example, leave the customizations at their defaults.

Enter a name and description for your task and then click **Run**.

You can see the status of your job on the **My Jobs** page.



Next steps

Congratulations on completing the tasks in this guide!

There is much more that you can do with IICS. Please check out the Cloud Data Integration documentation in the Informatica documentation portal to learn more.